

bit/bytes

↳ data storage
units

1 Kbyte $\rightarrow 10^3$ bytes

↓
images / light weighted
files.

$\{10^3 \text{ } \underline{\text{Mbytes}}\}$

↳ Bit

✓ ↓ ✓
 $\{ \underline{0} \quad \underline{1} \}$

Quantum

↳ Quantum bit

✓
CPU

32 bits $\rightarrow$ int

8 bits $\rightarrow$ byte.

$\{ \text{long} \}$

8 bytes $\rightarrow 8 * 8 \rightarrow \underline{\underline{64 \text{ bits}}}$

1 Mbytes $\rightarrow ? 10^6$ bytes.

$\{10^3 \text{ } \underline{\text{Mbytes}}\} \leftarrow 1 \text{ Gigabyte} \rightarrow 10^9 \text{ bytes.}$

$$(15)_{10} = (\dots)_2$$

$$\xrightarrow{\text{int}} \rightarrow \underline{\underline{32}}_{10}$$

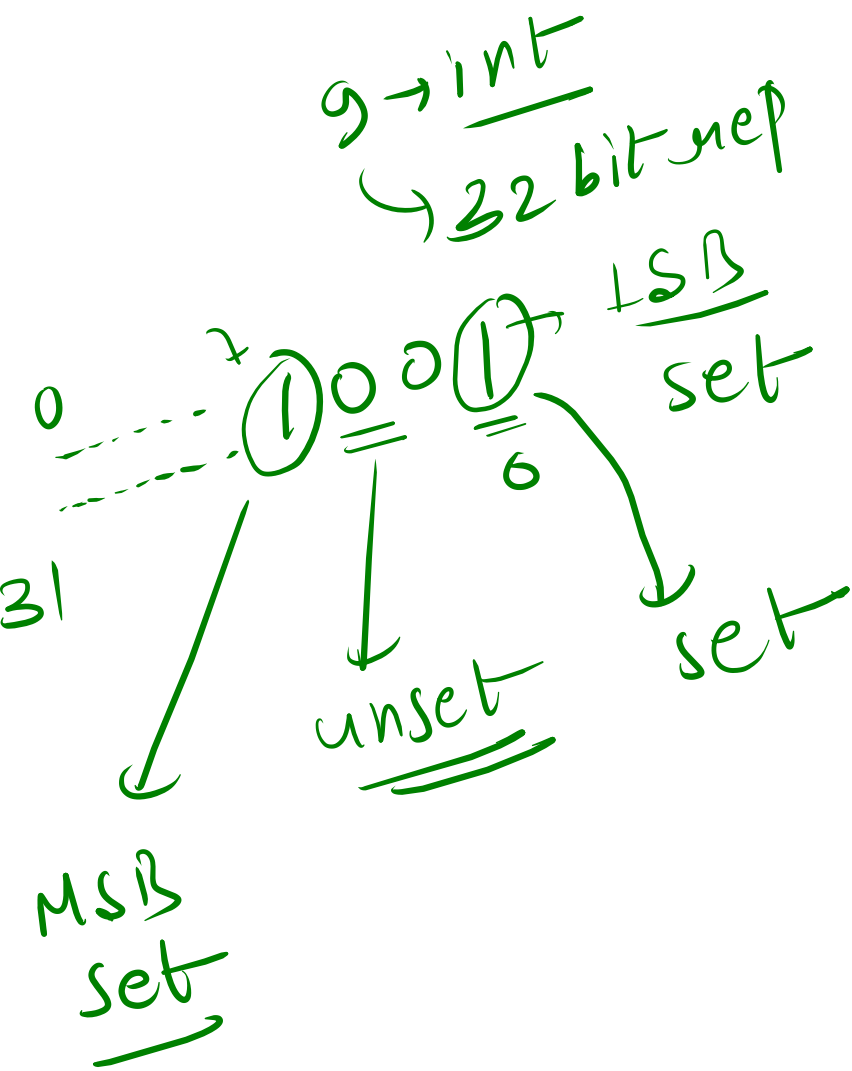
binary format } bit format

01...

$$\rightarrow \underline{\underline{1}}000000)_2$$

binary rep.?

2	32	✓✓
2	16	0
2	8	0
2	4	0
2	2	0
2	1	0
	0	1✓✓



2	15
2	7 1
2	3 1
2	1 1
	0 1

$$(1111)_2$$

$$1 * 2^3 + 0 * 2^2 + 0 * 2^1 + 1 * 2^0$$

$$(9)_{10} = (1001)_2$$

$$1 \rightarrow ? 0/3 \rightarrow 2^3 + 2^0 = 9$$

$$0 \rightarrow ? 1/2 \rightarrow 2^1 + 2^2 \otimes$$

num = 18 $\rightarrow$ how many

set bits

the no. is having.

2	18	
2	9	<u>0</u> $\rightarrow$
2	4	1 $\rightarrow +1$
2	2	0
2	1	0
2	0	1 $\rightarrow +1$

(X)

$\frac{0}{2} \dots 0$

$$② \simeq_{18} \rightarrow (10010) / 2 \rightarrow \underline{0}$$

$$9 \rightarrow (1001) / 2 \rightarrow \underline{1}$$

$$(10)_2 \xrightarrow{/2} (4) \leftarrow (100) / 2 = 0$$

$$\begin{aligned} & /2 = 1000 \\ & \text{reduced} \rightarrow (100) \\ & /2 \end{aligned}$$

$$(10) / 2 = 0$$

$$(1) / 2 = 1 + 1$$

$$\downarrow /2$$

$$(10)$$

$$/2 \rightarrow (1000)$$

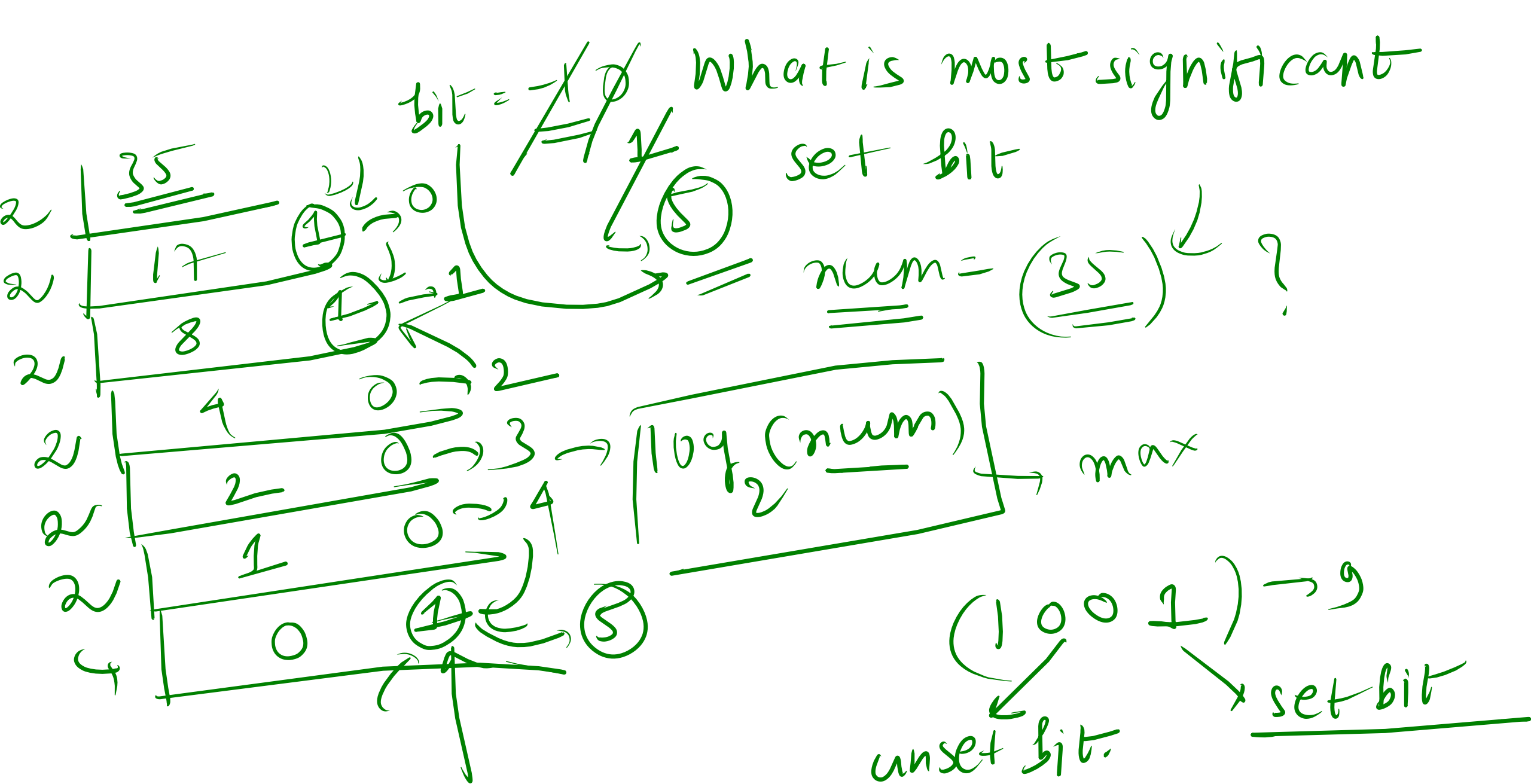
eight shift of n.

② bits

$$/2 = (1111)$$

$$\downarrow$$

$$\underline{\underline{RSOP.}}$$



$$11\underline{0} \Rightarrow (11)$$

$$\{6 \gg = 1\} \rightarrow \text{faster}$$

$$\underline{\underline{6/2 = 3^2}}$$

$$15 \rightarrow 11\underline{1}\cancel{1}$$

$$15 \gg = 1 \rightarrow (111)_2$$

$$15/2 \quad (\underline{111}\cancel{1})$$

$$\downarrow \quad 7 \rightarrow (111)$$

$$\gg = 1/2$$

$$\ll = 1 \rightarrow \underline{\underline{*2}}$$

$$\text{num} \gg = 1;$$

$$15 \ll = 1;$$

$$\downarrow \quad (1111)_2$$

$$\rightarrow 2^4 + 2^3 + 2^2 + 2^1$$

$$\underline{1100}_2$$

$$16 + 8 + 4 + 2$$

$$\downarrow \quad \underline{\underline{30}}$$

$$\underline{\underline{28}} >> = 3 ?$$

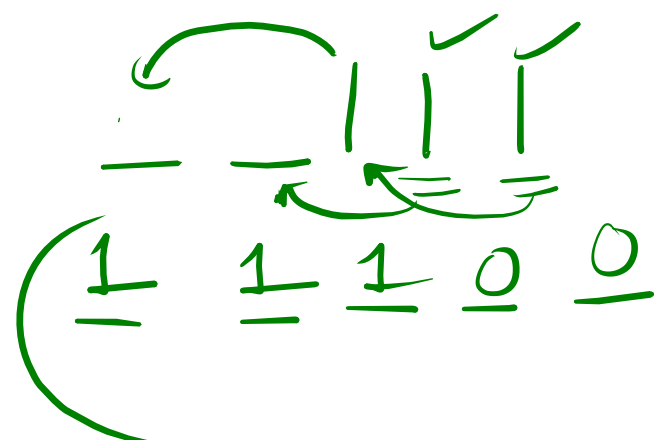

$$\begin{array}{ccccc} 1 & 1 & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 1 & 1 \\ \hline \end{array}$$

$$\downarrow = =$$

$$2 + 1 = 3.$$

$$7 \rightarrow (111)_2$$

$$7 << = (2) ?$$

$$\begin{array}{ccccc} 1 & 1 & 1 & 0 & 0 \\ \hline 1 & 1 & 1 & 0 & 0 \\ \hline \end{array}_2 = 2^4 + 2^3 + 2^2$$

$$= 16 + 8 + 4$$

$$= \underline{\underline{28}}$$

$$\begin{cases} \text{num} \gg x \rightarrow \text{num} / 2^x \\ \text{num} \ll x \rightarrow \text{num} * 2^x \end{cases}$$

bitwise operators $\rightarrow$ $\&$ $|$ $\wedge$ $\sim$
 $\hookrightarrow$ 2's complement

a & b
 $\downarrow$
 bitwise
 op.

$$\begin{array}{r} a = \underline{1} \underline{0} \underline{1} \underline{0} \\ \& b = \underline{1} \underline{1} \underline{0} \underline{0} \\ \hline \text{res} = (1 \ 0 \ 0 \ 0) = \end{array}$$

$$\& \rightarrow 1 \& 1 = 1$$

$$\left. \begin{array}{cc} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{array} \right\} = 0$$

$$\begin{array}{r}
 1010 \\
 \underline{1100} \\
 1110
 \end{array}$$

result $\leftarrow 1110$

$$\begin{array}{r}
 1010 \\
 \wedge 1100 \\
 \hline
 0110
 \end{array}$$

$$\begin{array}{r}
 11 \\
 11
 \end{array}
 \left. \vphantom{\begin{array}{r} 11 \\ 11 \end{array}} \right\} \textcircled{1}$$

$$0/0 \rightarrow \underline{0}$$

$$\begin{array}{r}
 110-1 \\
 0^11 \rightarrow 1
 \end{array}
 \left. \vphantom{\begin{array}{r} 110-1 \\ 0^11 \rightarrow 1 \end{array}} \right\}$$

$$\begin{array}{r}
 111-0 \\
 0^10 \rightarrow 1
 \end{array}$$

$1 \rightarrow \dots 00-1$
 $1 \ll i$
 $0001000\dots$

{ 2nd bit unset
 ↓
 set

$\dots 1 \overset{\uparrow}{0} 0 1 \equiv \underline{\underline{1101}}$
 $\begin{array}{r} \dots 0 \underline{1} 0 0 \\ \hline 1 \textcircled{1} 0 1 \end{array} \rightarrow \underline{\underline{\text{try}}}$

ith

$1 \ll \underline{i}$

$\dots \underline{00} 1 \ll 2 \rightarrow$
 $\dots 0100$

$\textcircled{1} =$
 $00 \dots \textcircled{1}$
 $\uparrow$

int

$$\begin{array}{r} 0 \dots 0 \underline{0} \underline{1} \underline{0} \underline{0} \underline{0} \underline{1} \underline{0} \\ 10 \dots \underline{0} \underline{0} \underline{1} \underline{0} \underline{0} \underline{0} \underline{0} \\ \hline 0 \dots 0 \underline{1} \underline{1} \underline{0} \underline{1} \underline{0} \end{array}$$

$$\begin{array}{c} 1 \ll 3 \\ \hline \end{array}$$

$$\begin{array}{c} \uparrow \quad \downarrow \\ \text{num} \quad (num) \mid (1 \ll i) \end{array}$$

set opr.

$$\cancel{10} \textcircled{1} 0 = \underline{\underline{1000}}$$

$$\downarrow \quad \quad \quad \downarrow$$

$$10 \textcircled{1} 1 = 1001$$

$$\begin{array}{r} 10 \textcircled{1} 1 \\ \times 00 \textcircled{1} 0 \\ \hline 0010 \oplus \end{array}$$

$$\begin{array}{r} 10 \quad 11 \\ \hline \downarrow \quad \downarrow \downarrow \\ 1001 \end{array}$$

$$\begin{array}{cccc} x & y & z & \\ \hline 1 & 1 & 0 & 1 \\ \hline \downarrow & \downarrow & \downarrow & \downarrow \\ x & y & z & \end{array}$$

bit

$$1 \& \underline{\underline{a}} = \underline{\underline{a}}$$

$$1 \& \underline{0} = 0$$

$$1 \& 1 = 1$$

ith bit $\rightarrow \underline{\underline{0}}$
other bits $\rightarrow 1$

$$\begin{array}{r} 1011 \\ \& 1101 \\ \hline 1001 \end{array}$$

$$\begin{array}{c} \sim(1 \ll i) \\ \downarrow \\ \sim 0010 \\ 1101 \end{array}$$

$$num = (num) \& (\sim(1 \ll i)) =$$

$\hookrightarrow$ unset opr.

num = 10²101

10001

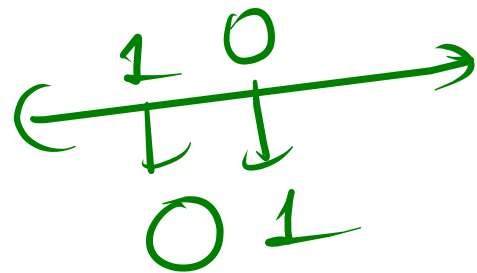
2 (11011) ₂

10001

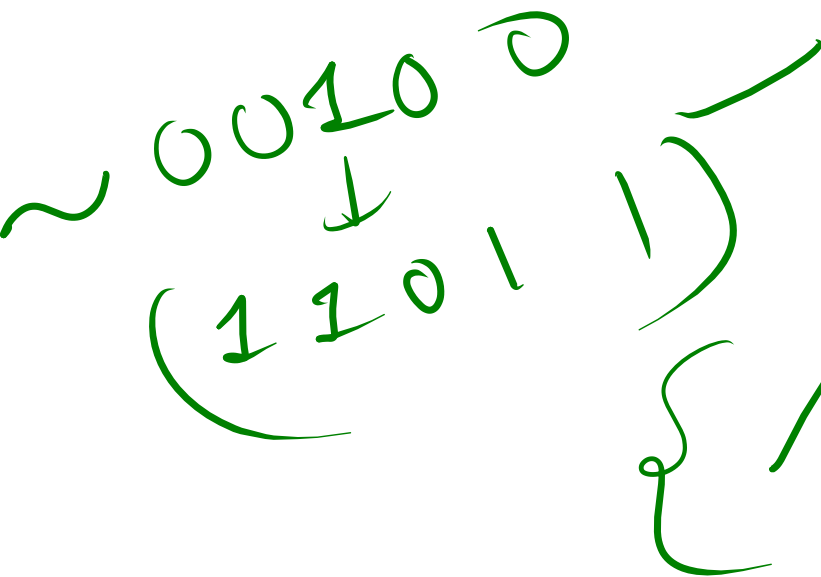
1 < 2

~(00100)

(11011)

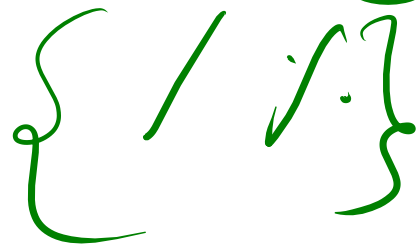


$num \& (\sim (1 \ll i))$



count the set bits

using bitwise ops.



+1

+1

+1

→ 3 set bit

$$n = (1010\textcircled{1})$$
$$\begin{array}{r} \& 0000\textcircled{1} \\ \hline 0000\textcircled{1} \end{array}$$

$$(n \& 1) = 1$$

10101

↓
1010~~1~~ → 0

↓
10~~1~~ → 1

↓
10~~1~~ → 0

1~~1~~ = 1

10~~1~~

count = 0;

while (n > 0) {

if ($n \& 1 == 1$) $\Rightarrow$ $\log(n)$

count++;

n >= 1;

$\rightarrow n/2 \leftarrow$

}

$n \wedge n$... odd $\rightarrow \underline{\underline{n}}$

$n \wedge n$... even $\rightarrow 0$

$n \rightarrow$

$$\underline{\underline{n \wedge n = ?}}$$

$n = 5 \rightarrow$

$$\begin{array}{r} \quad \quad \quad \underline{1} \underline{0} \underline{1} \\ \wedge \quad \underline{1} \underline{0} \underline{1} \\ \hline \quad \quad \underline{0} \underline{0} \underline{0} \\ \vee \quad \underline{1} \underline{1} \underline{0} \underline{1} \\ \hline \end{array}$$

$$\underline{1} \underline{0} \underline{1} \rightarrow (2)$$

$$\begin{array}{l} n \wedge n = 0 \\ (n \wedge n) \wedge n ? \\ \downarrow \\ 0 \wedge n \\ 0 \end{array}$$

arr → n

$x = arr[i]$

$e \times 2$
exc

$O(n)$
→ TC

x → arr
→ i ...
occ. once

arr → n numbers

→ odd

0 0 5

2 2 1 5 1 3 3

$x = 0 \wedge 2 \wedge 2 \wedge 1 \wedge 5 \wedge 1 \wedge 3 \wedge 3$

$x \rightarrow 0 \wedge 5 = 5$ → exceptional no.

$$\begin{array}{r}
 \rightarrow 1100 \overset{\curvearrowright}{1} \\
 \times 0000 \underline{1} \\
 \hline
 \text{res} \rightarrow 0000 \underline{1}
 \end{array}
 \quad
 \begin{array}{r}
 1 \gg 1 \\
 \downarrow \\
 \boxed{0} =
 \end{array}$$

$$\begin{array}{r}
 1100 \\
 0001 \\
 \hline
 0000
 \end{array}
 \quad
 \begin{array}{r}
 110 \\
 \times 001 \\
 \hline
 000
 \end{array}
 \quad
 \begin{array}{r}
 11 \\
 \times 01 \\
 \hline
 01
 \end{array}$$

$$(1100 \overset{\curvearrowright}{1}) \times \underline{1} = 1 \quad ? \quad + \underline{1}$$

$$\downarrow \gg 1$$

$$(110 \overset{\curvearrowright}{0}) \times 1 = 0 \quad ? \quad + 0$$

$$\downarrow \gg 1$$

$$(110) \times 1 = 0 \quad ? \quad + 0 \rightarrow \textcircled{3}$$

$$\downarrow \gg 1$$

$$(11) \times 1 = 1 \quad ? \quad + \underline{1}$$

$$\downarrow \gg 1$$

$$\underline{1 \gg 1} \quad 1 \times 1 \rightarrow 1 \quad ? \rightarrow + \underline{1}$$

$$\gg 1 \leadsto \sqrt{2} \rightarrow \log(n) \ll$$